

Master Thesis

for

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Scalable Point Cloud Attribute Compression

Skalierbare Kompression von Punktwolkenattributen

Point cloud attribute compression is essential for immersive communication, AR/VR, digital humans, and 3D scene representation, where color or reflectance information often accounts for a significant part of the total bitrate. Scalable compression is particularly valuable for bandwidth-adaptive transmission, as it enables a decoder to reconstruct point clouds at multiple quality levels from a single embedded bitstream, adapting to varying network conditions or application requirements. Despite recent advances in learning-based point cloud compression, scalable coding has been predominantly explored in the context of geometry, while scalable compression of point cloud attributes remains largely underinvestigated.

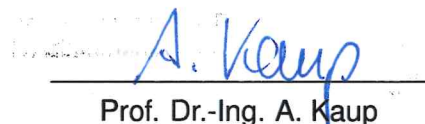
To address this gap, the objective of this thesis is to develop a scalable learning-based point cloud attribute compression framework, building upon Unicorn Part II: Attribute. Prior work has demonstrated that multiscale coding structures naturally lend themselves to scalable point cloud geometry compression, enabling multiple quality levels to be decoded from a single bitstream. Unicorn Part II: Attribute extends this paradigm to attribute coding, offering a multiscale framework that supports both lossless and lossy reconstruction. However, the framework is not explicitly designed for scalable attribute decoding. This thesis therefore investigates how to extend the multiscale attribute coding framework of Unicorn to support scalable point cloud attribute compression, with the goal of enabling progressive attribute reconstruction from a single bitstream while maintaining competitive rate-distortion performance.

Within the scope of this project, Mr. Tan will: (1) implement a baseline version of the Unicorn Part II attribute coding framework, (2) extend the attribute coding architecture to support scalable attribute compression.

A structured implementation and detailed documentation of the algorithms and experiments are part of the work.

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